

Indrani Sen

(206)458-2884 | indrani.sen1@gmail.com | Austin, TX | [LinkedIn](#) | H4 EAD (No sponsorship needed)

Results-driven Technical Program Manager with 6+ years of experience **and a recent MS in Computer Science (Georgia Tech, 2025)**. Specialized in leading distributed, scalable AI-enabled infrastructure and tooling initiatives across government and enterprise domains. Skilled in end-to-end program delivery, cloud migration, and data-driven automation. Committed to inclusive innovation and leveraging GenAI for operational excellence.

EDUCATION & CERTIFICATION

Georgia Institute of Technology | MS, Computer Science

Dec 2025

Kalinga Institute of Industrial Technology | MS, Computer Application

May 2014

AWS Certified Solutions Architect - Associate

Pursuing

PROFESSIONAL EXPERIENCE

Technical Program Manager, Tata Consultancy Services | Dec 2014 – Jun 2021

- **Led end-to-end technical programs** for utility and energy sector clients overall budget exceeding **\$2M**; designed and deployed AI-powered tools for workflow automation, including intelligent notification systems, managing complex stakeholder requirements and regulatory compliance across multiple state utilities
- **Led infrastructure modernization** for distributed utility systems, managing real-time data transmission across multi-state communication networks, coordinating hardware procurement, vendor integrations, and achieving **99%** uptime
- **Drove technical decision-making** for distributed processing architectures, managing cross-functional consensus on performance, power, and cost trade-offs that impacted overall system specifications
- **Delivered customer and partner relationship management** for utility clients, presenting technical solutions and program status to executive stakeholders while managing expectations and scope changes
- **Owned program-level cost and schedule commitments**, tracking \$2M+ budgets against technical milestones and delivering measurable performance improvements (40% response time, 14% resource optimization)
- **Coordinated cross-functional teams** of 15+ engineers using Agile methodologies and project management platforms (JIRA, GitLab CI), ensuring on-time delivery of infrastructure projects with complex dependencies.
- **Implemented cost optimization** delivering **\$200K** annual savings through supply chain efficiency improvements and vendor consolidation

TECHNICAL SKILLS

- **Program Management & Infrastructure:** Technical Project Management, Customer / Partner Relationship Management, Cross-functional Technical Leadership, JIRA, GitLab CI, Maven, Agile/Scrum Leadership
- **Data Center & Cloud Technologies:** Distributed Systems, Network Infrastructure, Industrial Communication Protocols, Enterprise Architecture, Container Orchestration (Docker), CI/CD Automation, Infrastructure Monitoring, 24/7 Operations Support
- **Enterprise Integration & APIs:** RESTful Services, Enterprise Integration Patterns, JWT Authentication, Service-Oriented Architecture, Microservices, API Management, Network Communication Systems

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- **Software Development & Tooling:** Java, Spring Boot, Python, Angular, React, System Integration Testing, Enterprise Software Development, Production Deployment Management
- **Database & Visualization:** MySQL, PostgreSQL, Oracle, Enterprise Data Integration, Amazon QuickSight, Tableau
- **AWS & Cloud Architecture:** AWS Solutions Architecture, Advanced Internet Computing, Cloud Migration, Scalable System Design

KEY TECHNICAL PROJECTS

GenAI Fairness Agent & RAG System – Independent Research | Dec 2025

- **Developed an AI Agent** using **RAG architecture** and Anthropic LLMs to audit employee performance reviews against **EEOC legal guidelines** and mitigate litigation risks
- **Built NLP pipelines** using **Sentence Transformers** and **FAISS** to detect gender-coded language patterns (agentic vs. communal) and calculate Disparate Impact metrics in real-time.
- **Designed an automated remediation workflow** that flags biased feedback and generates neutralized rewrites, preserving performance data while ensuring regulatory compliance

AWS Content Aggregation Platform – Personal Project

- **Developed enterprise-grade** serverless content processing system using Lambda, S3, Bedrock, and API Gateway
- **Implemented AI-powered categorization** with 90%+ accuracy and automated quality scoring
- **Built real-time monitoring** through CloudWatch with 3-second response times
- **Architected scalable solution** supporting 1000+ concurrent users with JWT authentication
- **Demonstrated end-to-end automation** from content submission to AI-driven analytics

GenAI-Powered Healthcare Risk Platform - Georgia Tech | 2025

- **Architected a provider-facing dashboard** using **FastAPI** (backend) and **React** (frontend), processing **FHIR-compliant** JSON patient data to calculate clinical risk scores
- **Integrated Google Gemini LLM** to transform Framingham risk assessments into personalized, evidence-based patient education and lifestyle recommendations
- **Managed end-to-end delivery** including **Docker** containerization for consistent deployment and the generation of synthetic datasets to test high-risk patient scenarios

PUBLICATIONS

"Path to Personalization: A Systematic Review of GenAI in Engineering Education" (KDD 2024)
Co-authored systematic review of 162 GenAI studies (KDD 2024); **led global team of 6 researchers** to define taxonomy and crowdsource content analysis.